NCBI Bookshelf. A service of the National Library of Medicine, National Institutes of Health.

StatPearls [Internet]. Treasure Island (FL): StatPearls Publishing; 2026 Jan-.

Nitroglycerin

Kyle H. Kim; Ghufran Adnan; Derek J. Schaller.

Author Information and Affiliations

Last Update: July 31, 2023.

Continuing Education Activity

Nitroglycerin is a vasodilatory drug used primarily to provide relief from anginal chest pain. It is currently FDA approved for the acute relief of an attack or acute prophylaxis of angina pectoris secondary to coronary artery disease. Off-label, non-FDA-approved uses include treatment for hypertensive urgency/emergency, coronary artery spasm, angina secondary to cocaine use, congestive heart failure (CHF), and chronic anal fissures. This activity outlines the indications, mechanism of action, methods of administration, important adverse effects, contraindications, monitoring, and toxicity of nitroglycerin, so providers can direct patient therapy successfully in instances where nitroglycerin benefits patient care.

Objectives:

- Identify the approved and off-label indications for using nitroglycerin.
- Summarize the mechanism of action of nitroglycerin.
- Outline nitroglycerin's adverse event profile.
- Describe interprofessional team strategies for improving care coordination and communication to properly use nitroglycerin to improve patient outcomes in the varied scenarios where it can be effective.

Access free multiple choice questions on this topic.

Indications

Nitroglycerin is a vasodilatory drug used primarily to provide relief from anginal chest pain. Nitroglycerin has been FDA approved since 2000 and was first sold as a brand-name agent. It is currently FDA approved for the acute relief of an attack or acute prophylaxis of angina pectoris secondary to coronary artery disease. Off-label, non-FDA-approved uses include treatment for hypertensive urgency/emergency, coronary artery spasm, angina secondary to cocaine use, congestive heart failure (CHF), and chronic anal fissures. Along with hydralazine as a combination therapy, it is also indicated in patients with heart failure with reduced ejection in whom ACE inhibitors are contraindicated. Nitroglycerin is occasionally given to reduce radial spasms during coronary angiography through the radial approach. Nitroglycerin was first used in 1879 as a treatment for angina and since it has been part of management to relieve anginal chest pain.[1]

Although nitroglycerin has a vasodilatory effect in both arteries and veins, the profound desired effects caused by nitroglycerin are primarily due to venodilation.[2] Venodilation causes pooling of blood within the venous system, reducing preload to the heart, which causes a decrease in cardiac work, reducing anginal symptoms secondary to demand ischemia. Arterial vasodilation

will still occur and contribute to the relief of anginal symptoms.[3][4] Vasodilation of the coronary arteries will cause increased blood flow to the heart, increasing perfusion, but this effect remains minimal compared to the effects of venodilation.

Mechanism of Action

As with other nitrates used to treat anginal chest pain, nitroglycerin converts to nitric oxide (NO) in the body. NO then activates the enzyme guanylyl cyclase, which converts guanosine triphosphate (GTP) to guanosine 3',5'-monophosphate (cGMP) in vascular smooth muscle and other tissues. cGMP then activates many protein kinase-dependent phosphorylations, which enhances the reuptake of calcium into the sarcoplasmic reticulum, increases extracellular calcium, and opens the calcium-gated potassium channel.[5][6] This ultimately results in the dephosphorylation of myosin light chains within smooth muscle fibers.[7] This activity causes the relaxation of smooth muscle within blood vessels, resulting in the desired vasodilatory effect.

Administration

Nitroglycerin is most commonly administered as a tablet that is absorbed sublingually. It is given in hospitals as well as prescribed for outpatient use. Patients may be prescribed nitroglycerin as prophylaxis for anginal chest pain prior to an event that may provoke anginal symptoms. They must be instructed to allow the nitroglycerin to dissolve in their mouth and allow their oral mucosa to absorb the drug. There currently are three doses available: 0.3 mg, 0.4 mg, and 0.6 mg. The dose is repeatable every 5 minutes until the achievement of relief. If anginal pain persists after three doses, prompt medical attention is required. After administration, the onset of vasodilatory effects occurs within 1 to 3 minutes, with a max effect occurring within 5 minutes. Nitroglycerin is primarily eliminated via metabolism in the liver and has a mean half-life of approximately 2 to 3 minutes.[8]

There are intravenous (IV) routes of administration for nitroglycerin used most commonly in emergency rooms and intensive care units (ICU). It is administered as a 5% dextrose in drip and is indicated when sublingual nitroglycerin has failed to provide symptomatic relief or if rapid and continued relief of symptoms is necessary. Intravenous nitroglycerin is frequently used to treat acute coronary syndromes, hypertensive emergency, and acute congestive heart failure (CHF) exacerbations.[9][10] When administered, its effect requires tight vitals monitoring, as discussed below. It should be kept in mind that continuous intravenous infusion of nitrates can lead to the development of tolerance. It is important to switch patients to oral form with intermittent dose and long nitrate-free intervals to avoid tolerance. Sometimes in a catheterization laboratory, nitroglycerin can be given intracoronary as a treatment for no-reflow.[11] Nitroglycerin provides rapid coronary vasodilation when given intracoronary.

Transdermal methods of nitroglycerin administration are also commonly administered in emergency rooms during acute angina attacks. It is a 2% ointment applied directly to the patient's skin and allowed to absorb. It is typically for patients that cannot tolerate the sublingual administration of nitroglycerin or have a previous adverse reaction to the sublingual tablet. Absorption takes about 5 to 10 minutes for full effect.[12][13] The ideal application is on a surface with minimal hair as hair can inhibit absorption, and care is necessary not to apply repeatedly to the same area if multiple doses are required. Repeated application of the ointment to the same area can cause skin irritation and dermatitis. Transdermal patches are also available at varying doses per hour (0.1 mg, 0.2 mg, 0.4 mg, and 0.6 mg), but these are rarely used and

reserved for angina prophylaxis only. Similar application precautions as apply to the ointment also apply to the patches.

Adverse Effects

Nitroglycerin has many adverse effects of significance, most resulting from the vasodilatory effects of the medication.[14][15] These include:

- Dizziness
- Weakness
- Palpitations
- Vertigo
- Headaches
- Nausea
- Vomiting
- Diaphoresis
- Syncope

Many of these adverse effects are secondary to the hypotensive effects of nitroglycerin. Patients may report symptoms of orthostatic hypotension, which manifest as dizziness, weakness, palpitations, and vertigo. Profound hypotension may occur in patients with preload-dependent conditions. Some patients can be more sensitive to the hypotension caused by nitrates, potentially resulting in nausea, vomiting, diaphoresis, pallor, and collapse even at therapeutic doses. There have also been reports of flushing, exfoliative dermatitis, and drug rash in some patients taking nitroglycerin. Headaches can be severe, throbbing, and persistent and may occur immediately after use.

Syncope is the most dangerous adverse effect and can result in falls and their resultant injuries. The risk of syncope significantly increases with the concurrent use of a phosphodiesterase-5 (PDE-5) inhibitor. A further discussion is included below under contraindications.[16] In patients with right coronary artery occlusion and chest pain, right ventricular infarction should be ruled out before giving nitroglycerin as nitroglycerin causes venodilation and reduces preload. The right ventricle is volume dependent chamber, and reduction in preload can cause profound hypotension, leading to cardiogenic shock.[17]

Contraindications

The contraindications of nitroglycerine therapy include:

- Allergic reactions to nitroglycerin are extremely rare, but reports do exist. Nitroglycerin is contraindicated in patients that have reported allergic symptoms to the medication.[18]
- Known history of increased intracranial pressure, severe anemia, right-sided myocardial infarction, or hypersensitivity to nitroglycerin are contraindications to nitroglycerin therapy.

- Concurrent use of nitroglycerin with PDE-5 inhibitors (e.g., sildenafil citrate, vardenafil hydroxide, tadalafil) is absolutely contraindicated. PDE-5 inhibitors have proven to accentuate the hypotensive effects of nitrates and precipitate syncopal episodes.[19]

Monitoring

Any testing does not currently monitor nitroglycerin levels as its half-life is approximately 2 to 3 minutes, and the drug undergoes rapid metabolism in the liver. When administered intravenously in the emergency room or ICU, its effects are often very closely monitored for real-time blood pressure monitoring. This vigilance is necessary to maximize the effectiveness and provide rapid feedback on the patient's condition. When administered sublingually, nitroglycerin's effectiveness is typically measured via the resolution of symptoms; angina, hypertension, and congestive heart failure. In the event of overdose, monitoring vital signs may be necessary to monitor the hemodynamic effects of nitroglycerin. Continuous monitoring of blood pressure, heart rate, respiratory rate, and oxygen saturation is recommended.[20][21]

Nitroglycerin is a protein-bound drug, and it undergoes hepatic metabolism. Therefore it has numerous drug interactions. Before prescribing, providers should determine if the patient is taking any medications that may interact with nitroglycerin. Common interactions include alteplase, heparin, tricyclic antidepressants, and other anticholinergic drugs. Alcohol intake should also be limited. Nitroglycerin is pregnancy category C, and its use requires caution in breastfeeding mothers. It is not currently known whether nitroglycerin is excreted in breast milk.

Toxicity

Overdose toxicity from nitroglycerin is mainly a consequence of increased vasodilatory response. Hypotension, venous pooling, increased vasodilation, and reduced cardiac output can be expected in these patients. Compensatory effects, such as tachycardia and palpitations, can also be expected. Vasodilation and venous pooling can increase the amount of blood in the cranial space, resulting in increased intracranial pressures; this can cause persistent, throbbing headaches, along with confusion, fever, vertigo, nausea, vomiting, and visual disturbances.[22] As intracranial pressure increases, symptoms will progress to dyspnea secondary to a reduced respiratory effort, heart block, bradycardia, paralysis, seizures, coma, and, eventually, death.

No currently known antagonist is available to counteract the effect of nitroglycerin. Since the effects are related to venodilation and relative arterial hypovolemia, efforts to increase central fluid volume have proven effective. Intravenous administration of normal saline, in addition to the passive elevation of the patient's legs, may provide adequate support, but there are no controlled trials to prove its effectiveness. Epinephrine or other arterial vasoconstricting agents are not recommended as they will not likely improve the patient's condition and may cause more difficulties in the future. Methemoglobinemia has some rare reports as a consequence of nitrate overdose. Clinician suspicion should be raised in patients with hypoxemia symptoms despite a lack of respiratory symptoms, normal arterial PO2, and adequate cardiac output. Blood from patients with methemoglobinemia has a "chocolate brown" appearance in color, with no change in color upon exposure to air.[23] The treatment for methemoglobinemia is an intravenous administration of methylene blue, dosed at 1 to 2 mg/kg of the patient's body weight.

Enhancing Healthcare Team Outcomes

Interprofessional team members including clinicians, cardiology specialists, primary care providers, pharmacists, internists, and nursing staff who work with patients taking nitroglycerin should be fully aware of the indications and contraindications of the drug, as well as the potential adverse effects. Overall, nitroglycerin is relatively safe, and monitoring of the levels is not required. In the event of overdose, monitoring vital signs may be necessary to monitor the hemodynamic effects of nitroglycerin. Continuous monitoring of blood pressure, heart rate, respiratory rate, and oxygen saturation is recommended. Pharmacists can assist the team by monitoring for drug-drug interactions. Nitroglycerin is pregnancy category C, and its use requires caution in breastfeeding mothers. It is currently unknown whether nitroglycerin is excreted in breast milk. Interprofessional collaboration and information sharing will drive better outcomes with fewer adverse events with nitroglycerine therapy. [Level 5]

Review Questions

- Access free multiple choice questions on this topic.
- Comment on this article.

References

1. Parratt JR. Nitroglycerin--the first one hundred years: new facts about an old drug. J Pharm Pharmacol. 1979 Dec;31(12):801-9. [PubMed: 43362]
2. Arnold WP, Mittal CK, Katsuki S, Murad F. Nitric oxide activates guanylate cyclase and increases guanosine 3':5'-cyclic monophosphate levels in various tissue preparations. Proc Natl Acad Sci U S A. 1977 Aug;74(8):3203-7. [PMC free article: PMC431498] [PubMed: 20623]
3. Thadani U, Lipicky RJ. Short and long-acting oral nitrates for stable angina pectoris. Cardiovasc Drugs Ther. 1994 Aug;8(4):611-23. [PubMed: 7848896]
4. Brown BG, Bolson E, Petersen RB, Pierce CD, Dodge HT. The mechanisms of nitroglycerin action: stenosis vasodilatation as a major component of the drug response. Circulation. 1981 Dec;64(6):1089-97. [PubMed: 6794931]
5. Carvajal JA, Germain AM, Huidobro-Toro JP, Weiner CP. Molecular mechanism of cGMP-mediated smooth muscle relaxation. J Cell Physiol. 2000 Sep;184(3):409-20. [PubMed: 10911373]
6. Zhao Y, Vanhoutte PM, Leung SW. Vascular nitric oxide: Beyond eNOS. J Pharmacol Sci. 2015 Oct;129(2):83-94. [PubMed: 26499181]
7. Webb RC. Smooth muscle contraction and relaxation. Adv Physiol Educ. 2003 Dec;27(1-4):201-6. [PubMed: 14627618]
8. Boden WE, Padala SK, Cabral KP, Buschmann IR, Sidhu MS. Role of short-acting nitroglycerin in the management of ischemic heart disease. Drug Des Devel Ther. 2015;9:4793-805. [PMC free article: PMC4548722] [PubMed: 26316714]
9. Holt DB, Pang PS. Vasodilator Therapies in the Treatment of Acute Heart Failure. Curr Heart Fail Rep. 2019 Feb;16(1):32-37. [PubMed: 30762175]
10. Rhoney D, Peacock WF. Intravenous therapy for hypertensive emergencies, part 1. Am J Health Syst Pharm. 2009 Aug 01;66(15):1343-52. [PubMed: 19635770]
11. Piana RN, Paik GY, Moscucci M, Cohen DJ, Gibson CM, Kugelmass AD, Carrozza JP, Kuntz RE, Baim DS. Incidence and treatment of 'no-reflow' after percutaneous coronary intervention. Circulation. 1994 Jun;89(6):2514-8. [PubMed: 8205658]
12.

Todd PA, Goa KL, Langtry HD. Transdermal nitroglycerin (glyceryl trinitrate). A review of its pharmacology and therapeutic use. Drugs. 1990 Dec;40(6):880-902. [PubMed: 2127741]

13. Taylor SH. The role of transdermal nitroglycerin in the treatment of coronary heart disease. Am Heart J. 1986 Jul;112(1):197-207. [PubMed: 3088957]
14. Divakaran S, Loscalzo J. The Role of Nitroglycerin and Other Nitrogen Oxides in Cardiovascular Therapeutics. J Am Coll Cardiol. 2017 Nov 07;70(19):2393-2410. [PMC free article: PMC5687289] [PubMed: 29096811]
15. Drugs and Lactation Database (LactMed®) [Internet]. National Institute of Child Health and Human Development; Bethesda (MD): Oct 15, 2024. Nitroglycerin. [PubMed: 30000576]
16. Kloner RA. Pharmacology and drug interaction effects of the phosphodiesterase 5 inhibitors: focus on alpha-blocker interactions. Am J Cardiol. 2005 Dec 26;96(12B):42M-46M. [PubMed: 16387566]
17. Ferguson JJ, Diver DJ, Boldt M, Pasternak RC. Significance of nitroglycerin-induced hypotension with inferior wall acute myocardial infarction. Am J Cardiol. 1989 Aug 01;64(5):311-4. [PubMed: 2502902]
18. Ramey JT, Lockey RF. Allergic and nonallergic reactions to nitroglycerin. Allergy Asthma Proc. 2006 May-Jun;27(3):273-80. [PubMed: 16913273]
19. Ishikura F, Beppu S, Hamada T, Khandheria BK, Seward JB, Nehra A. Effects of sildenafil citrate (Viagra) combined with nitrate on the heart. Circulation. 2000 Nov 14;102(20):2516-21. [PubMed: 11076826]
20. Bosson N, Isakson B, Morgan JA, Kaji AH, Uner A, Hurley K, Henry TD, Niemann JT. Safety and Effectiveness of Field Nitroglycerin in Patients with Suspected ST Elevation Myocardial Infarction. Prehosp Emerg Care. 2019 Sep-Oct;23(5):603-611. [PubMed: 30556765]
21. Long B, Koyfman A, Gottlieb M. Management of Heart Failure in the Emergency Department Setting: An Evidence-Based Review of the Literature. J Emerg Med. 2018 Nov;55(5):635-646. [PubMed: 30266198]
22. Ghani GA, Sung YF, Weinstein MS, Tindall GT, Fleischer AS. Effects of intravenous nitroglycerin on the intracranial pressure and volume pressure response. J Neurosurg. 1983 Apr;58(4):562-5. [PubMed: 6402569]
23. Buenger JW, Mauro VF. Organic nitrate-induced methemoglobinemia. DICP. 1989 Apr;23(4):283-8. [PubMed: 2658373]

Disclosure: Kyle Kim declares no relevant financial relationships with ineligible companies.

Disclosure: Ghufran Adnan declares no relevant financial relationships with ineligible companies.

Disclosure: Derek Schaller declares no relevant financial relationships with ineligible companies.

Copyright © 2026, StatPearls Publishing LLC.

This book is distributed under the terms of the Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International (CC BY-NC-ND 4.0) (http://creativecommons.org/licenses/by-nc-nd/4.0/), which permits others to distribute the work, provided that the article is not altered or used commercially. You are not required to obtain permission to distribute this article, provided that you credit the author and journal.

Bookshelf ID: NBK482382 PMID: 29494004